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PRE PSPM
DP024

2022/2023

Instruction: This question paper consists of 6 subjective questions.
Attempt all questions.

QUESTION	MARKS
1	/ 17
2	/ 12
3	/ 9
4	/ 14
5	/ 14
6	/ 14
TOTAL	/ 80

LIST OF FORMULA

1. $x(t) = Asin\omega t$
2. $\omega = \frac{2\pi}{T} = 2\pi f$
3. $k = \frac{2\pi}{\lambda}$
4. $v = f\lambda$
5. $y(x,t) = Asin(\omega t \pm kx)$
6. $C = \frac{Q}{V}$
7. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$
8. $C_{eff} = C_1 + C_2 + \cdots + C_n$
9. $\tau = RC$
10. $Q = Q_o e^{-\frac{t}{RC}}$
11. $Q = Q_o\left(1 - e^{-\frac{t}{RC}}\right)$
12. $I = \frac{dQ}{dt}$
13. $Q = ne$
14. $V = IR$
15. $\rho = \frac{RA}{l}$
16. $R_{eff} = R_1 + R_2 + \cdots R_n$
17. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$
18. $B = \frac{\mu_o I}{2\pi r}$
19. $B = \frac{\mu_o NI}{2R}$
20. $B = \mu_o nI$
21. $\phi_B = BAcos\theta$
22. $\Phi_B = N\phi_B$
23. $\varepsilon = -\frac{d\phi_B}{dt}$
24. $\varepsilon = Blvsin\theta$
25. $\varepsilon = -NA\frac{dB}{dt}$
26. $\varepsilon = -NB\frac{dA}{dt}$
27. $R = 2f$
28. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
29. $m = -\frac{v}{u}$
30. $m = -\frac{h_i}{h_o}$
31. $y_m = \frac{m\lambda D}{d}$
32. $y_m = \frac{(m+\frac{1}{2})\lambda D}{d}$
33. $\Delta y = \frac{\lambda D}{d}$

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	=	$3.00\times10^{8}\,\mathrm{m\,s^{-1}}$
Permeability constant	μ_0	=	$4\pi\times10^{-7}\,\mathrm{H\,m^{-1}}$
Elementary charge	e	=	$1.60\times10^{-19}\,C$
Electron mass	m_e	=	$9.11\times10^{-31}\,kg$
Neutron mass	m_n	=	$1.67\times10^{-27}\,kg$
Proton mass	m_p	=	$1.67\times10^{-27}\,kg$
Deuteron mass	m_d	=	$3.34\times10^{-27}\,kg$
Avogadro constant	N_A	=	$6.02\times10^{23}\,\mathrm{mol^{-1}}$
Free-fall acceleration	g	=	$9.81\ \mathrm{m\,s^{-2}}$
Density of water	ρ_w	=	$1000kg\,m^{-3}$ Pa

1.(a) A progressive wave is given by equation $y = 4.0sin(90t - 0.2x)\pi$, where t is in second, x is in cm and y is in m.

(i) What is the direction of propagation of the wave?

(ii) What is the wavelength of the wave in meters?

(iii) What is the distance propagated by the wave after 60 seconds?

[7 marks]

1. (b) A progressive wave equation is given as $y = 0.2\sin\left(200\pi t - \frac{\pi x}{15}\right)$, where t is in second, x is in cm and y is in cm. Determine:

(i) velocity of the wave,

(ii) maximum velocity of vibrating particles,

(iii) velocity of a particle, located 453 cm from source of the wave at time t = 0.05 s,

(iv) maximum acceleration of vibrating particles.

[10 marks]

2. (a) A 100 Ampere of current flows through the filament of a light bulb. Determine the number of free electrons flowing through the filament in 10 minutes. *[3 marks]*

2.(b) (i) Calculate the resistance per unit length of a Nichrome wire, which has a radius of 0.321 mm.
(Given resistivity of Nichrome is $1.5 \times 10^{-6}\ \Omega$m)

(ii) If the potential difference of 10 V is maintained across a 1.0 m length of the Nichrome wire, what is the current in the wire?

[4 marks]

2. (c)(i)

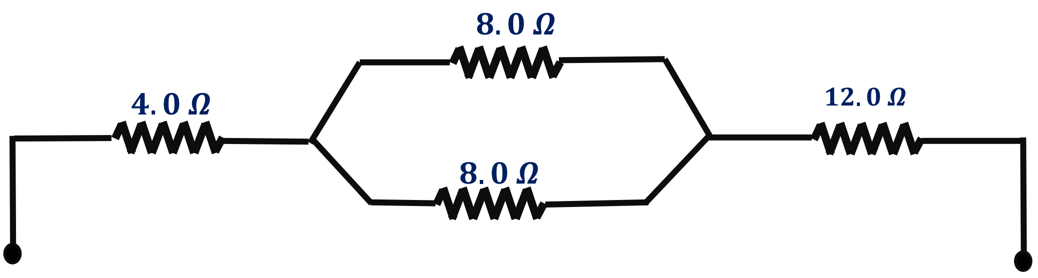

FIGURE 1

Find the equivalent resistance for the circuit as shown in FIGURE 1.

(ii) If potential difference of 40 volt is applied across the circuit, determine potential difference across the resistor of $12.0\ \Omega$.

[5 marks]

3. (a)

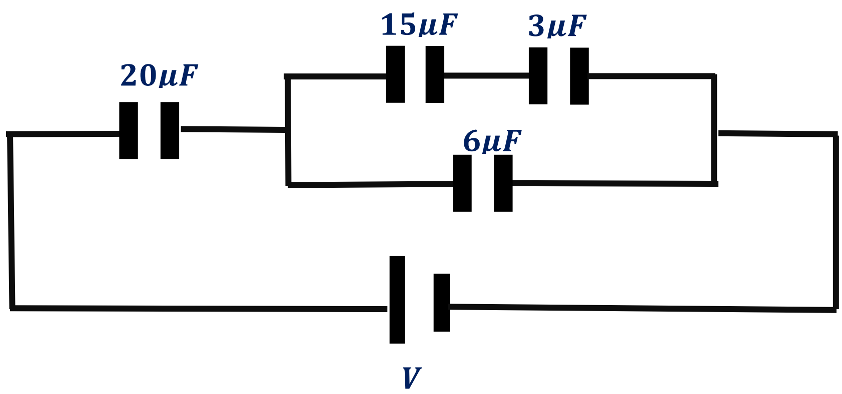

FIGURE 2

Four capacitors are arranged as shown in the FIGURE 2.

(i) What is the effective capacitance of the circuit ?

(ii) If V is 18 volt, determine the amount of charge supplied by the battery to the circuit. *[5 marks]*

3. (b) A fully charged 12.0 μF capacitor has 6.0 mC charge on each plate. The capacitor is then discharged through a 1 MΩ resistor.

(i) Calculate the time constant for discharging circuit.

(ii) Calculate the time taken if just 20% of charge is left in the capacitor.

[4 marks]

4. (a)

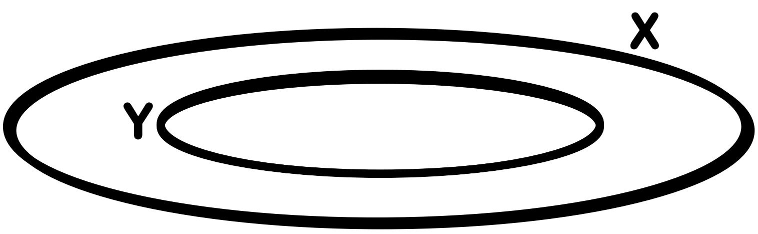

FIGURE 3

Two flat circular coils X and Y carrying the same current of 5.0 A are arranged coaxially. Coil X has 200 turns and radius 10.0 cm. Coil Y has 300 turns and radius 5.0 cm. Calculate the magnitude and direction of resultant magnetic flux density at centre of the coils if the currents in the coils are flowing in clockwise direction.

[5 marks]

4(b)

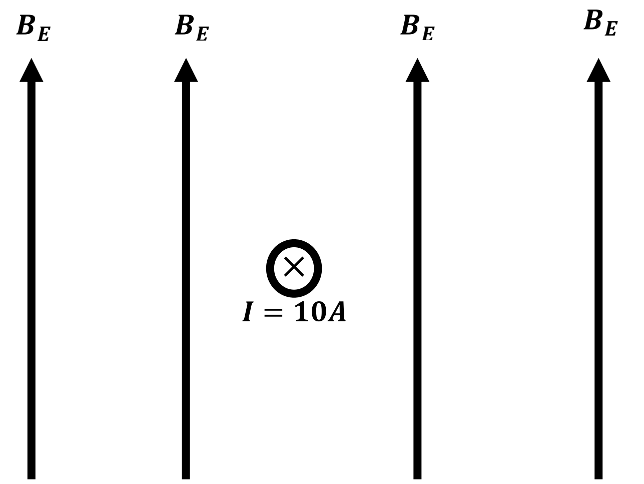

FIGURE 4
(TOP VIEW)

The top view figure above shows a long straight wire carries current of 10 A. If magnitude of the Earth's magnetic field , B_E is $2.0 \times 10^{-5} T$, Calculate the magnitude of resultant magnetic at a distance 0.10 m to the west of the wire . ***[3 marks]***

4(c)

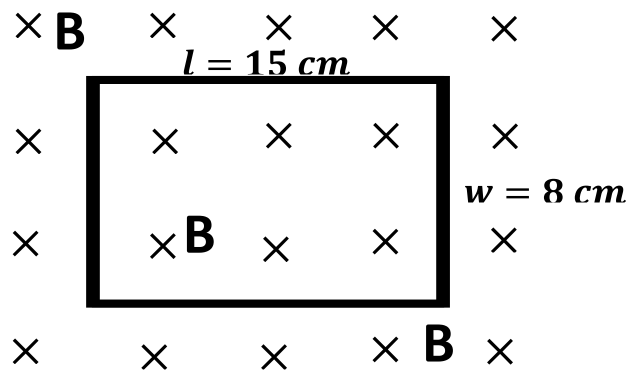

FIGURE 5

Figure 5 shows a rectangular coil is placed perpendicularly with uniform magnetic field B. The coil has 45 turns and the magnitude of the magnetic field B is 0.25 T. The length , l of the coil suddenly decreases from 15 cm to 10 cm within 0.8 second.

(i) What is the magnitude of induced emf triggered when the length of the coil , l is decreasing ?

(ii) If the coil has resistance of $0.4\ \Omega$. What is the magnitude of induced current flowing through the coil and also its direction?

[6 marks]

5(a). A vertical object is placed in front of a spherical mirror and the linear magnification of the image formed is 3. If an upright image is formed 30.0 cm from the vertex of the mirror,

(i) state the type of spherical mirror,

(ii) calculate the distance of the object from the pole of the mirror,

(iii) calculate the radius of curvature of the mirror. ***[6 marks]***

5(b). A lens forms an upright image of an object placed at a distance 5.0cm from lens. The image formed is three times bigger than the object.

(i) State the type of lens used.

(ii) Calculate the focal length of the lens.

(iii) Calculate the distance between the object and the image formed.

(iv) If the image formed is real and with the same magnification, determine the new position of object. ***[8 marks]***

6(a). Two narrow slits are 0.025 mm apart. When a laser shines on them, a series of fringes forms on a screen that is a meter away. These fringes are 3.0 cm apart. What is the separation between the second order bright fringe and the central bright?

[4 marks]

6(b). A monochromatic light source of wavelength 656 nm is used in a double slit experiment. The slits separation, d = 0.5 mm and the screen is located 1.2 m from the slits.

(i) Determine the distance between third order dark fringes on both sides of central bright.

(ii) What is the separation between 2 consecutive dark fringes?

[5 marks]

6(c). In a double slit experiment, the distance between the screen and the slits is 2.0 m. The distance between the center of the interference pattern and tenth bright fringe is 3.4 cm.

(i) If the wavelength of the light used is 5.0×10^{-7} m, determine the slit separation.

(ii) Calculate the separation between 8^{th} dark fringe and 4^{th} bright fringe.

[5 marks]

SOALAN TAMAT